DOT Net : Assignment-3

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Course - B.Tech (SE B, 6th Sem
Subject - DOT Net

1. Explain the steps involved in creating and managing a windows Service.

Soln A window service is a long-running background application that runs without a user interface

Steps to create a Window Service (C#):

1. Create Project
   - Open visual Studio
   - Select: windows Service (.NET Framework) or Worker Service (.NET)
2. Define Service class
   - Inherit from ServiceBase
   - Override methods:
     - OnStart()
     - OnStop()
3. Write Service Logic
   - Add background tasks inside OnStart()
   - Use threads/timers if needed.
4. Add Installer
   - Add Project Installer
   - Configure ServiceProcessInstaller and ServiceInstaller
5. Build the Project
   - Compile to generate .exe
6. Install the Service

   Use Command Prompt:

   ```
   installutil YourService.exe
   ```

   Or use:

   ```
   sc create ServiceName binPath="path"
   ```
7. Start/Stop Service

   Use:
   - services.msc
   - Or command:

   ```
   net start ServiceName
   net stop ServiceName
   ```

Managing Windows Service:

- Start / Stop / Pause / Resume using Service Control Manager
- Monitor logs using Event Viewer
- Update Service by reinstalling updated version

2. Describe the process of creating a windows forms application using Standard Controls.

Soln: windows forms is used to create desktop GUI applications in C#.

Steps:

1. Create Project
   - Open Visual Studio
   - Select: Windows Forms App
2. Design Form
   - Drag and drop controls from Toolbox:
     - button
     - label
     - TextBox
     - ComboBox, etc.
3. Set Properties
   - Change properties like:
     - Name
     - Text
     - Size
     - BackColor
4. Write Event Code
   - Double-click control (e.g., button) to generate event handler.
5. Run Application
   - Press F5

Example

```
private void button1_Click(Object sender, EventArgs e) {
        label1.Text = "Hello world";
}
```

3. What are event handlers? Explain how events are handled in Windows Forms.

Soln: Event Handler:

An event handler is a method that executes when a specific event occurs (like button_click, mouse move, keypress)

nt Handling in Windows Forms:

indows forms uses delegates and events.

Steps:

1. User performs an action (click button)
2. Event is triggered (click event)
3. Event handler method is executed.

Example

```
private void button1_Click(object sender, EventArgs e) {
        MessageBox.Show("Button clicked!");
}
```

Types of Events:

- Click
- MouseHover
- KeyPress
- Load.

Ways to attach event:

- Designer (double click)
- Code:

```
button1.Click += button1_Click;
```

- Explain MDI forms and their advantages in GUI applications

Soln MDI

- It stands for Multiple Document Interface.
- It allows multiple child windows inside a single parent window.

Components:

- MDI Parent form → Main container
- MDI Child form → forms inside parent.

How to create MDI:

1. set parent form property:

```
IsMdiContainer = true;
```

2. create child form:

```
Form2 f = new Form2();
f.MdiParent = this;
f.Show();
```

Advantages:

- multiple documents in one window
- better organization of UI
- easy navigation b/w forms.
- saves desktop space
- common in applications like MS word (old versions)

5. Explain XML handling in C# using:

a. XML DOM
b. XMLWriter
c. XML Reader

Soln XML is used to store and transfer structured data

a) XML DOM (Document Object Model)

- It loads entire XML into memory as a tree structure
- It allows read, modify, delete nodes

features:

- easy navigation
- supports modification
- memory heavy.

b) XMLWriter

- used to write XML files sequentially
- fast and efficient

features:

- fast writing
- forward-only
- cannot modify existing XML

c) XML Reader

- reads XML data in forward-only mode
- lightweight and fast

features:

- fast reading
- low memory usage
- cannot modify XML

```
Document doc = new XmlDocument();
doc.Load("data.xml");
XmlNode node = doc.SelectSingleNode("//student/name");
Console.WriteLine(node.InnerText);
```

XML DOM

```
XmlWriter writer = XmlWriter.Create("data.xml");
writer.WriteStartDocument();
writer.WriteStartElement("student");

writer.WriteElementString("name", "John");
writer.WriteElementString("age", "20");

writer.WriteEndElement();
writer.WriteEndDocument();

writer.Close();
```

XML Writer

```
XmlReader reader = XmlReader.Create("data.xml");
While (reader.Read()){
    if (reader.NodeType == XmlNodeType.Element){
        Console.WriteLine (reader.Name);
    }
}
reader.Close();
```

XML Reader